

Omar Mahmoud

Data Engineer | Machine Learning Engineer

omarmahmoudzamalkawy@gmail.com | +201155579144 | Cairo, Egypt | LinkedIn | GitHub | Portfolio

Professional Summary

Computer Science student at Helwan University specializing in Data Engineering through the Digital Egypt Pioneers Initiative (DEPI). Experienced in designing and implementing ETL/ELT pipelines, data preprocessing, SQL optimization, and data visualization using Power BI and Tableau. Proficient in Python, SQL, Apache Spark, Apache Airflow, and Apache Kafka with hands-on experience developing machine learning pipelines for predictive analytics and data-driven solutions.

Education

Bachelor of Computer Science · *Helwan University*

Aug 2024 – Jul 2028

Cairo, Egypt

GPA: 3.1 / 4.0 (Very Good) Degree in Progress

Experience

Data Engineer Intern · *Digital Egyptian Pioneers Initiative (DEPI)*

Nov 2025 – Jul 2026

Cairo, Egypt

- Designed and maintained ETL/ELT pipelines for analytics and enterprise reporting workflows.
- Cleaned, transformed, and integrated data using Python (Pandas) and advanced SQL queries.
- Optimized relational database schemas to improve query performance and reduce latency.
- Applied data modeling, orchestration (Apache Airflow), and data quality best practices.

Certifications

Data Engineering – AI & Data Science · *Digital Egypt Pioneers Initiative (DEPI)* Jul 2026

[View Certificate]

ETL pipelines, data processing, data analysis, and machine learning fundamentals using industry-standard tools.

Machine Learning · *National Telecommunication Institute (NTI)* [View Certificate]

July 2026

Supervised and unsupervised learning, model training, evaluation techniques, and ML pipelines.

Projects

Customer Churn Prediction Pipeline

Dec 2025 – May 2026

Python, LightGBM, SQL Server, Streamlit, Scikit-learn, Pandas

- Led a 5-person team to build an end-to-end ML pipeline predicting telecom customer churn on 999,999 records and 32 features (DEPI Graduation Project).
- Engineered a full preprocessing pipeline: median imputation, ordinal & one-hot encoding, StandardScaler, and stratified 80/20 train-test split across 30 features.
- Trained and compared 4 classification models (LightGBM, Logistic Regression, XGBoost, Random Forest) with `class_weight='balanced'` to handle 9:1 class imbalance; LightGBM selected with ROC-AUC 0.635 and 63.6% churn recall.
- Loaded dataset into Microsoft SQL Server and executed 5 analytical queries revealing key drivers: Month-to-Month contracts at 26.5% churn vs. 5.6% for Two-Year contracts.
- Deployed an interactive 3-page Streamlit dashboard with KPI cards, EDA charts, and a real-time prediction tool with auto-detected risk factors. [Live Demo]

Monitor Home System

Mar 2026 – May 2026

PHP, MySQL, HTML, CSS, JavaScript

- Developed a full-stack home monitoring web application with real-time data processing.
- Designed backend services for managing home system data and implemented CRUD functionalities.
- Built responsive user interfaces with modern UI/UX design practices.

Helwan Wings – Airport Management System

Nov 2025 – Jan 2026

Java, SQL, UML

- Designed an airport management and ticket booking system with full database integration.
- Created system analysis diagrams including UML and database schema designs.
- Implemented modules for flight scheduling, passenger management, and ticket booking.

Hyper-Market System

Nov 2025 – Jan 2026

Java, OOP

- Built a Java console application using OOP principles to manage products, stock, and categories.
- Implemented inventory tracking and product management to improve operational efficiency.

Fashion Shop – E-Commerce Frontend

Apr 2025 – May 2026

HTML, CSS, JavaScript

- Developed a responsive e-commerce website with product filtering and shopping cart management.
- Improved UX through responsive layouts and modern front-end design practices.

Technical Skills

Programming Languages	Python (Pandas, NumPy, Matplotlib), SQL, Java, Data Structures & Algorithms
Databases	MySQL, PostgreSQL, MongoDB, Microsoft SQL Server, Database Design & Optimization
Data Engineering & ETL	Apache Spark, Apache Airflow, Apache Kafka, Hadoop, ETL/ELT Pipelines, Azure Data Factory
Cloud & DevOps	Microsoft Azure, Docker, Git & GitHub
Data Visualization	Power BI, Jupyter Notebook
Machine Learning	Scikit-learn, LightGBM, XGBoost, Model Evaluation, Class Imbalance Handling

Languages
Arabic – Native English – Proficient